Anthony James Padavano

BACKEND ENGINEER — ENTERPRISE PLATFORM & IDENTITY

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LINKS

Portfolio · GitHub · agentic-titan · ORGANVM System

PROFILE

Designed and operates an 8-organization platform coordinating 113 repositories through a machine-readable registry, promotion state machine (LOCAL→CANDIDATE→PUBLIC_PROCESS→GRADUATED→ARCHIVED), and automated dependency validation enforcing 50 directed edges across org boundaries with 0 violations. 2,650 Python source files; 23,470 tests (14,500 Python/pytest). Authentication and identity patterns across multi-org governance: role-based access controls, seed.yaml contracts defining produces/consumes edges per repo, and a pulse daemon computing system density every 15 minutes. The infrastructure that lets one person ship at the coherence of a team — precisely the scale problem LangChain's enterprise customers face.

TECHNICAL SKILLS

Python · FastAPI · pytest (14,500+ tests) · API Design · Authentication & Identity Patterns · Multi-Org Governance · Promotion State Machines · Registry Architecture · Dependency Validation · MCP Protocol · CI/CD · GitHub Actions · PostgreSQL · Neon · TypeScript · Next.js · LaunchAgent Automation · YAML Schema Design · Documentation Architecture (739K words)

SELECTED PROJECTS

ORGANVM Registry & Promotion State Machine — 113-Repo Enterprise Governance
Designed registry-v2.json (2,240-line single source of truth across 8 GitHub orgs), seed.yaml contracts (produces/consumes edges per repo), and a promotion state machine enforcing sequential transitions (LOCAL→CANDIDATE→PUBLIC_PROCESS→GRADUATED→ARCHIVED). 50 dependency edges validated automatically; 0 violations in production. Directly analogous to LangSmith's deployment lifecycle and LangGraph's graph-state management for enterprise-scale agent pipelines.

organvm-mcp-server — 88-Tool Identity & Access Layer
MCP server (stdio transport) exposing 88 tools across 16 permission groups: registry queries, seed contract inspection, dependency tracing, session management, governance enforcement, and coordination primitives. Role-differentiated access patterns: read-only graph tools vs. write-capable governance tools vs. coordination primitives. Python; strict schema validation on all inputs. Enables AI agents to operate within defined authorization boundaries without filesystem access.

system-dashboard — FastAPI Observability Platform
FastAPI + Jinja2 + HTMX dashboard with live system health, registry browser, dependency graph visualization, soak monitoring, and omega scorecard. 739K-word documentation corpus served via structured routes (/health/, /registry/, /graph/, /soak/, /essays/, /omega/). Auto-refreshes from registry on each request; no stale cache.

Application Pipeline — Python CLI Tooling Suite with 23,470 Tests
Python CLI infrastructure: 9-dimension scoring rubric, YAML pipeline state machine with forward-only progression enforcement, campaign orchestrator, submission preflight validation, batch enrichment, ATS API clients (Greenhouse, Lever, Ashby), and multi-model IRA diagnostic system. 2,650 Python source files; 23,470 tests (14,500 pytest). Modular architecture: 6 decomposed library modules share common foundation via backward-compatible re-exports.

stakeholder-portal — Next.js + Neon Intelligence Portal
Next.js stakeholder portal surfacing live registry state to non-technical audiences. TypeScript strict mode; Neon (PostgreSQL) backend; Vercel deployment. Authentication layer separating public browsing from privileged governance actions.

EXPERIENCE

Independent Backend Engineer & Platform Architect — 2020 — Present

- **Enterprise Governance Infrastructure:** Designed promotion state machine (LOCAL→CANDIDATE→PUBLIC_PROCESS→GRADUATED→ARCHIVED) across 113 repos in 8 GitHub organizations; automated dependency validation enforcing 50 directed edges with 0 violations.
- **Python Backend at Scale:** 2,650 Python source files; 23,470 total tests (14,500 pytest); modular library architecture with 6 decomposed modules and backward-compatible re-exports. FastAPI services with strict schema validation.
- **Identity & Access Patterns:** Role-differentiated MCP server (88 tools, 16 permission groups) enforcing authorization boundaries for AI agent sessions; seed.yaml contracts as machine-readable access control declarations per repository.
- **Documentation Architecture:** 739K-word documentation corpus across 8 organs; structured routes, registry-driven auto-refresh, and canonical identity propagation from single source of truth to all downstream consumers.

Instructor — Composition & Writing — 2015 — Present

Miami Dade College, Keiser University, Nova Southeastern, FAU, Young Harris, Tarrant County, Broward, Palm Beach State

- 100+ courses across 8 institutions; 2,000+ students. Documentation, technical communication, structured argumentation at scale.

Digital Marketing Manager — 2023 — 2024

Miami Dade College Foundation (nonprofit)

- Rebuilt foundation website; designed campaigns increasing donor engagement by 32%.

Multimedia Specialist — 2011 — 2020

AJP Media Arts — clients incl. Boca Ballet Theatre, Forward Funding, Arts in Boca

- Video, audio, and visual campaigns for nonprofits and arts organizations; 17.5M+ views, $2M in fundraising.

EDUCATION

M.F.A. Creative Writing, Florida Atlantic University — 2015 — 2018

B.A. English Literature, CUNY College of Staten Island — 2010 — 2014

CERTIFICATIONS

Full-Stack Developer, Meta (2024) · UX Design, Google (2023) · Digital Marketing, Google (2023) · Project Management, Google (2023)